

Proof skeleton

Exact Diffusion Design for Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks

Topology-Wide All-Spectrum Localization and Exponent-Optimal Stationary Heterogeneity Trade-Offs

1. Reaction algebra and topology-wide localization

Construct Y_m and Γ_m directly from the indexed reactions. Reaction-by-reaction verification gives $c^T \Gamma_m = 0$ for $c = (0, 4, \dots, 4, 2, 1)^T$. An explicit $m \times m$ minor proves $\text{rank } \Gamma_m = m$, while solving the balance equations yields

$$\ker \Gamma_m = \text{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

Therefore every positive flux is $(a\mathbf{1}_m, b, b)$. Differentiation at the unit equilibrium gives $A_m(a, b)$; at a general equilibrium $x^* = H^{-1}\mathbf{1}$, the Jacobian is $A_m(a, b)H$. Conversely $k_r = v_r/(x^*)^{y_r}$ realizes every $a, b, H > 0$.

For a principal set with fewer than m vertices, Frobenius decomposition reduces every nontrivial SCC to: the $X_1, \dots, X_{m-1}$ long cycle; the $X_2, \dots, X_m$ long cycle; a principal boundary-triad block; or a negative singleton. At $b = 2a$ one directed edge disappears and only refines the decomposition. The two cycle characteristic equations are excluded from $\Re \lambda \geq 0$ by strict product inequalities. The full triad has positive cubic coefficients and $c_1 c_2 - c_3$ is a sum of fourteen positive monomials. Every lower-order principal matrix is thus Hurwitz. Sparse expansion of the $X_1, \dots, X_m$ block gives the negative signed determinant $-2a^{m-1}b \prod h_i$, forcing a positive real eigenvalue.

Audit object. Independently generate every induced SCC for $m = 3, \dots, 8$ at generic b and at $b = 2a$, while the proof uses the all- m graph classification rather than enumeration.

2. Principal-minor diffusion-ray theorem and exact network law

For $a_I = (-1)^{|I|} \det J_{I,I}$ and $D \succ 0$, multilinearity gives

$$\det(sD - J) = \sum_I a_I \prod_{j \notin I} s d_j = s(\beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}).$$

Assume directly that $a_I > 0$ for $|I| \leq n-2$ and that $\sum_{|I|=n-1} a_I > 0$; a simple zero and otherwise stable spectrum supplies the latter coefficient condition in the application. Hence $\beta_k > 0$ for $k \geq 2$, and the bracket is strictly increasing on $s > 0$, proving the unique simple threshold iff $\beta_1 < 0$.

For the stronger positive-real band, define

$$\chi_s(\lambda) = \det(\lambda I + sD - J) = \sum_I a_I \prod_{j \notin I} (\lambda + s d_j).$$

Every derivative contribution from $|I| \leq n-2$ is nonnegative. The total order- $(n-1)$ derivative is $\sum_{|I|=n-1} a_I > 0$, the coefficient of λ in $\det(\lambda I - J)$. Hence $\chi'_s(\lambda) > 0$ for $\lambda \geq 0$. The sign of $\chi_s(0)$ therefore gives a positive real eigenvalue exactly for $0 < s < s_*$. At threshold, $\chi_{s_*}(0) = 0$ and $\chi'_{s_*}(0) > 0$; this, rather than scalar simplicity of the generalized-pencil root alone, proves ordinary algebraic simplicity of zero for $J - s_* D$.

For $A_m(a, b)H$, compute every order- m omission minor. Omitting Z gives weight -2 ; omitting an interior X_j gives $+16$; omitting X_1 or X_m gives zero. Thus

$$\beta_1 = 2a^{m-1}b \prod_{i=1}^m h_i \left(8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z \right).$$

This proves the exact criterion. The same sign follows from first-order perturbation of the conservation eigenvalue. For fixed H , minimizing $d_{\max}/d_{\min}$ under the strict inequality gives the nonattained infimum $T(H) = 8h_Z \sum_{j=2}^{m-1} h_j^{-1}$. At $H = I$, $T = 8(m-2)$, and $h_Z/h_j \geq 1/\chi_H$ gives $\chi_D \chi_H > 8(m-2)$.

3. Improved unit-equilibrium primary bifurcation

Set $a = b = 1$, $H = I$, $K_i = 91m - 181 - i$, and use

$$d_1 = 23/63, \quad d_i = K_i^{-1}, \quad d_m = 1/7, \quad d_Z = 16/45.$$

An explicit right vector satisfies $(A_m - D_m)r = 0$ and an explicit left vector satisfies $(A_m - D_m)^T \ell = 0$. Exact harmonic-sum formulas prove $\ell^T r < 0$ and $\ell^T D_m r < 0$, hence transversality $\eta_m > 0$.

Homogeneous stability follows from

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \quad \lambda q_m = (1 + \lambda)^{m-3} P(\lambda) - R(\lambda),$$

where $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. The exact difference $|1 + \lambda|^2 |P|^2 - |R|^2$ expands into 35 nonnegative monomials in $x = \Re \lambda$ and $y^2 = (\Im \lambda)^2$, with equality only at the origin. For $m = 3$, $q_3 = (\lambda + 7)(\lambda^2 + 5\lambda + 2)$ is Hurwitz. For $m \geq 4$, $|1 + \lambda| \geq 1$ makes the 35-term strict inequality incompatible with $(1 + \lambda)^{m-3} P = R$ at any nonzero closed-right-half-plane root. Since $q_m(0) = 16m - 34 > 0$, the conservation zero is simple.

For spatial damping tD_m , set $g_1 = \lambda + 2 + (23/63)t$, $g_m = \lambda + 5 + t/7$, $g_Z = \lambda + 4 + (16/45)t$, $F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z$, and $G = g_Z(4g_1 + g_m) - 36$. Chain elimination gives $\det(\lambda I - A_m + tD_m) = Q_m F - G$. At $(\lambda, t) = (0, 1)$, the interior product telescopes to $91/90$. With $z = y^2$ and $s = t - 1$, the exact source

$$E_{77} = [(91/90)^2 + z] |F(x + i\sqrt{z}, 1 + s)|^2 - |G(x + i\sqrt{z}, 1 + s)|^2$$

has 77 strictly positive coefficients and no constant term. Thus equality occurs only at $(x, z, s) = (0, 0, 0)$. Hence the first mode has one simple zero, all other first-mode eigenvalues are stable, and all $t > 1$ modes are Hurwitz.

The exact unit contrast is

$$\chi_D = \frac{23(91m - 183)}{63},$$

which improves the superseded high-contrast profile and is within a dimension-independent factor of the sharp linear threshold.

4. Conservation-compatible normal form and stable branch

The mass-action field is quadratic:

$$f(\mathbf{1} + u) = A_m u + \frac{1}{2} B_m(u, u), \quad c^T B_m(u, v) = 0.$$

On the fixed integrated-mass class, the critical mode is $r \cos \xi$. Since $\cos^2 \xi = (1 + \cos 2\xi)/2$, solve

$$A_m w_0 = -\frac{1}{4} B(r, r), \quad c^T w_0 = 0, \quad (A_m - 4D_m)w_2 = -\frac{1}{4} B(r, r).$$

The first is an augmented nonsingular solve; the second is invertible by mode isolation. The interior w_2 recurrence telescopes through four consecutive K_i factors, leaving an explicit 3×3 boundary system.

Projection onto the adjoint critical mode yields

$$\dot{A} = \eta_m \mu A + c_m A^3 + O(|A|^5 + |A|^3 |\mu| + |A| |\mu|^2),$$

$$c_m = \frac{\ell^T \{B(r, w_0) + \frac{1}{2} B(r, w_2)\}}{\ell^T r}.$$

Writing $\mathfrak{h}_m = \sum_{j=1}^{m-2} K_j^{-1}$, the numerator telescopes to $R_m + C_m \mathfrak{h}_m$. The supplement prints R_m , C_m , their positive denominators, and the exact L_m clearing identity. Shifted coefficientwise-positive polynomials prove the numerator positive, while $\ell^T r < 0$; therefore $c_m < 0$ for every $m \geq 3$. For a direct audit, put

$$\mathcal{Q}_m = 589180301m^3 - 3500015940m^2 + 6930529579m - 4574434500,$$

$$\begin{aligned} P_R &= 68605040480814208768m^4 - 550882186169626030957m^3 + 1658612632937449670852m^2 \\ &\quad - 2219226476204103501323m + 1113379274975809565700, \end{aligned}$$

$$\begin{aligned} P_C &= 652054120726848m^4 - 5151971981328467m^3 + 15265080924982572m^2 \\ &\quad - 20102347725659113m + 9927281930180400. \end{aligned}$$

Then

$$R_m = \frac{P_R}{286118780220(8m-17)\mathcal{Q}_m}, \quad C_m = -\frac{215P_C}{11645046(8m-17)\mathcal{Q}_m},$$

and exact clearing gives

$$R_m + C_m \frac{m-2}{90m-179} = \frac{L_m}{286118780220(8m-17)(90m-179)\mathcal{Q}_m}.$$

The shifted coefficients of $\mathcal{Q}_m, P_R, P_C, L_m$ are positive, so $C_m < 0$ and $R_m + C_m \mathfrak{h}_m \geq R_m + C_m(m-2)/(90m-179) > 0$.

Crandall–Rabinowitz gives two positive branches for $\mu > 0$. Their center eigenvalue is $-2\eta_m \mu + O(\mu^2) < 0$. The complementary spectrum is separated at onset and remains separated under analytic perturbation. Positive Neumann diffusion is sectorial on fixed-mass L^2 , its fractional H^1 domain is a Banach algebra in one dimension, and the quadratic Nemytskii map is smooth. The principle of linearized stability gives local exponential convergence in H^1 for sufficiently close nonnegative initial data in the same conservation class.

5. Equilibrium-scaled stable trade-off and audit boundaries

Let $\nu = m - 2$ and

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \geq 2, \end{cases} \quad L_0 = \frac{\kappa_\nu}{\sqrt{\nu}}, \quad L_1 = \frac{90\nu}{90\nu + 1}.$$

The interval is nonempty: for $\nu = 1$ the squared inequality is $8281 < 24300$, while for $\nu \geq 2$ it follows from $(90\nu + 1)^2 \leq (91\nu)^2 < 6480\nu^3$. Set $h_1 = h_m = h_Z = 1$, $h_i = K_i/(LK_{i-1})$, and choose physical diffusion $D^{\text{phys}} = H_m(L)\Delta$. Under $z = H_m(L)x$,

$$z_t = H_m(L)\{f(z) + \Delta z_{xx}\}.$$

The effective critical bracket and right vector are unchanged; the dynamic left vector is $\tilde{\ell}_m(L) = H_m(L)^{-1}\ell_m$. The physical fixed-mass functional is $(H_m(L)^{-1}c)^T z$.

Chain elimination now gives interior factors

$$g_i = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

For $t \geq 1$, Bernoulli's inequality and $\nu L^2 \geq 1/3$ reduce spatial half-plane exclusion to the exact source polynomial

$$E_{84} = (91/90)^2(1 + Ax + z/3)|F(x + i\sqrt{z}, 1 + s)|^2 - |G(x + i\sqrt{z}, 1 + s)|^2, \quad A = 2\nu L, z = y^2, s = t - 1.$$

Its 84 coefficients are coefficientwise nonnegative in $A > 0$, and equality occurs only at the first-mode zero $(x, z, s) = (0, 0, 0)$.

At $t = 0$, the exact homogeneous determinant, after dividing by $\det H_m(L)$, is

$$QF_0 - R, \quad Q = \prod_{i=2}^{m-1} \left(1 + L \frac{K_{i-1}}{K_i} \lambda\right),$$

$$F_0 = \lambda^3 + 11\lambda^2 + 31\lambda + 16, \quad R = 5\lambda^2 + 33\lambda + 16.$$

Every factor of Q has modulus at least $|1 + L\lambda|$ for $\Re \lambda = x \geq 0$. If $\nu \geq 2$, then

$$|Q|^2 \geq 1 + 2\nu Lx + \nu L^2 y^2 \geq 1 + Ax + \frac{5}{4}y^2, \quad A = 2\nu L.$$

On the certified domain $A \geq \sqrt{5\nu} \geq \sqrt{10} > 1/4$. With $z = y^2$ and $U = A - 1/4$, the exact source

$$E_{22} = [1 + (U + 1/4)x + (5/4)z]|F_0(x + i\sqrt{z})|^2 - |R(x + i\sqrt{z})|^2$$

has 22 nonzero monomials. All coefficient polynomials are nonnegative and equality is possible only at $\lambda = 0$. The coefficient $5/4$ is sharp for this certificate only: replacing it by B makes the $z = y^2$ coefficient at $x = 0$ equal to $256B - 320$. This does not identify an intrinsic dynamical boundary. Also

$$(QF_0 - R)'(0) = 16L \sum_{i=2}^{m-1} K_{i-1}/K_i - 2 > 0,$$

so the conservation zero is simple. If $\nu = 1$, putting $c = 91L/90$ gives

$$\frac{QF_0 - R}{\lambda} = c\lambda^3 + (1 + 11c)\lambda^2 + (6 + 31c)\lambda + (16c - 2),$$

whose coefficients and Routh–Hurwitz difference $6 + 99c + 325c^2$ are positive because $c > 1/8$.

The second harmonic is unchanged, while the zero harmonic changes by $w_0(L) = w_0^{\text{ref}} + \tau_m(L)\rho$. The cubic numerator is $N_m(L) = N_m^{\text{ref}} + \tau_m(L)S_m$. With $1/91 \leq \mathfrak{h}_m < 1/90$,

$$S_m = -\frac{4(1760850\mathfrak{h}_m - 10253)}{462105}, \quad -\frac{1}{10} < S_m < 0.$$

Since $C_m < 0$, exact clearing gives

$$N_m^{\text{ref}} - \frac{1}{100} \geq \frac{P_{\text{ref}}(\nu)}{D_{\text{ref}}(\nu)},$$

where

$$\begin{aligned}
P_{\text{ref}} &= 3790502986637265684840\nu^5 - 974216530468600286489\nu^4 - 53103567440921218871\nu^3 \\
&\quad - 576386186827093561\nu^2 + 3649732858601219\nu + 55281268032918, \\
D_{\text{ref}} &= 715296950550(8\nu - 1)(90\nu + 1) \\
&\quad \times (589180301\nu^3 + 35065866\nu^2 + 629431\nu + 3306).
\end{aligned}$$

$D_{\text{ref}} > 0$, and the shift $\nu = v + 1$ gives six positive coefficients for P_{ref} , proving $N_m^{\text{ref}} > 1/100$.

For the gauge, set

$$\begin{aligned}
A_\tau &= 1494249120\mathfrak{h}_m L\nu^2 - 69786990\mathfrak{h}_m L\nu + 108738630L\nu^2 \\
&\quad + 1214388L\nu - 8521L - 125249670\nu^2 + 1031940\nu, \\
B_\tau &= 32760\mathfrak{h}_m L\nu + 32760L\nu^2 + 4L - 4095\nu.
\end{aligned}$$

Then $\tau_m = -A_\tau/[15876(8\nu - 1)B_\tau]$ and $B_\tau = 4095\nu[8L(\nu + \mathfrak{h}_m) - 1] + 4L > 0$. The exact derivatives are negative on $L \geq 1/\sqrt{3\nu}$, so the maximum is at $(\mathfrak{h}_m, L) = (1/91, 1/\sqrt{3\nu})$. For $\nu \geq 2$ its endpoint sign is bounded by

$$P_{\text{up}}(\nu) = -\frac{189709065}{2}\nu^3 - 507201030\nu^2 - 935658\nu + 58481,$$

with

$$P'_{\text{up}} = -\frac{569127195}{2}\nu^2 - 1014402060\nu - 935658 < 0, \quad P_{\text{up}}(2) = -2789453215 < 0.$$

The case $\nu = 1$ uses the exact unreduced endpoint expression and $\sqrt{3} < 7/4$. Hence $\tau_m(L) < 1/20$. These bounds hold on the broader cubic range $L \geq 1/\sqrt{3\nu}$ and give $N_m(L) > 1/200$, hence $c_m(L) < 0$ throughout the certified interval.

The physical contrasts are

$$\chi_D = \frac{23}{63}91\nu L, \quad \chi_H = \frac{91\nu - 1}{91\nu L}.$$

Here χ_H measures equilibrium concentration-scale separation, not a universal measure of biological expense. Moreover

$$\chi_D(L)\chi_H(L) = \frac{23(91\nu - 1)}{63} = \frac{23(91m - 183)}{63} = \chi_D^{\text{unit}}(m).$$

Thus L reallocates a fixed product and reduces $\max(\chi_D, \chi_H)$, not the product. At L_0 ,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \quad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu} \frac{91\nu - 1}{91\nu},$$

so both are $\Theta(\sqrt{m})$. The topology-specific necessary bound for stationary crossings implies any such realization must have $\max(\chi_D, \chi_H) > \sqrt{8\nu}$, so exponent $1/2$ is optimal among stationary-crossing realizations of $\hat{N}_m$. Constants remain open.

Failure tests. Verify the SCC exhaustion at $b = 2a$; the principal-minor derivative argument; the 22/35/77/84-term equality cases and the direct $\nu = 1$ cubic; the transformed physical diffusion; the fixed-mass gauge; and the comparison $N_m(L) > 1/200$. No finite regression substitutes for these all- m identities.